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RESEARCH AND TECHNOLOGY INSTITUTE
LABORATORY OF DATA ENGINEERING

Web Application for DPCfam and DPCstruct Data Exploration

Report 4 (3 Feb - 8 Feb 2026) : Web Development
(DPCFam Backend)

Emmanuel Nyandu Kagarabi
Master in Data Management and Curation (MDMC - SISSA)

Supervisors :
Valerio Piomponi
Elaheh Saadat

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1 Introduction

This report is the fourth in a sequence describing the completion of weekly assigned tasks during our internship. Focusing on the DPCFam dataset(Russo et al., 2022), we present the backend methodology used to produce the second draft of the DPCFam web app, where, once on the global landing page, a user must choose DPCFam as the database, enter a valid Metacluster ID, and then, in case everything goes well, they can explore Metacluster properties, download some files (Seeds, HMM model, MSA), and inspect the AlphaFold properties and 3D model of their representative (if any). If a user has chosen PFam on the landing page and entered a valid ID, they can display a list of all matching DPCFam MCs and their average overlap. A local PostgreSQL database is used for development purposes. We've organised this report as a full workflow/README for reproducibility; other relevant aspects could be discussed during our demo session.

2 Development

2.1 Database

During the week, I updated our database to extend the DPCFam app: First, the `mcs_properties` table was altered by including two additional columns relevant to DPCFam (Figure 1):

	mcid	size	uniref50	avg_len	std_avg_len	lc_percent	cc_percent	dis_percent	tm	pfam_da	da_percent	size_pfam	avg_ov_percent	overlap_label
0	MC1	17931	185.68	28.77	4.72	0.00	18.44	0.01	PF13614	44.23	6332	80.82	equivalent	
1	MC4	617	59.91	6.07	4.99	0.00	1.87	1.26	PF03600	62.84	345	7.54	shifted	
2	MC15	139	81.21	5.05	4.96	0.18	13.81	0.03	UNKNOWN	0.00	131	0.00	NONE	
3	MC19	120	71.57	7.70	8.85	2.25	4.86	1.69	PF11915	94.07	111	13.66	shifted	
4	MC21	937	91.20	7.70	2.65	0.00	24.38	0.00	PF01012	98.90	809	34.52	shifted	

Figure 1: Full DPCFam properties : We have included `std_avg_len` and `avg_ovg_percent` columns for some needs in the DPCFam app. 46 828 MCIDs

Next, we created another table, `mcs_sequences`, containing information related to seeds per MCID as shown below (Figure 2):

	mcid	protein_id	seq_range	seq_length	aa_seq
0	MC82027	E8W9U6	2-131	130	ILLSTSDTDLSSARAAQGPVSYRYANPSRLSLDALPQLLEGTDLV...
1	MC82027	I7DYJ9	1-143	143	MHVVFRESHGLDESDTPQDLGQTPADLVVLSFSDSLGFAAGWHR...
2	MC82027	A0A0M6ZLI4	13-175	163	DGGDAVDLGQAPADILFLSAADTELGSFSAHSAALGTDASLRLAN...
3	MC82027	A0A1X4ITI5	1-182	182	MHLLAATPGNLDGPLEVDLGQSPADLVVLSAADTDLAALSEARNL...
4	MC82027	A0A1Q7WDG5	5-140	136	FVLLSTADTDLLAARASGAPWRVANPTRLGAADLPALLDGATLAVV...

Figure 2: Seeds : 15 401 952 samples for all MCIDs.

Lastly, we added `alphafold_reps`, containing information about representative of

MCs(if any) resulting in a file like Figure 3

	mcid	alphafold_prot	seq_range	hmm_coverage	avg_plddt
0	MC1	AF-P0A149-F1	4-181	94.12	96.24
1	MC4	AF-Q04671-F1	770-830	96.83	93.80
2	MC19	AF-A0A1C1CP96-F1	66-135	98.59	83.62
3	MC21	AF-P97089-F1	1-93	100.00	93.57
4	MC24	AF-P9WJB2-F1	161-250	97.78	89.73

Figure 3: AlphaFold representatives : 38 668 MCs, others are missing (like MC15).

For each table, a .csv file is available and is actually needed to create that table in PostgreSQL.

2.2 Workflow

We provide a full workflow/README allowing the display of the web app in its current state.

Prerequisites

We are currently using the following tools on Linux : Git (version 2.43.0), Python (version 3.12.3), PostgreSQL(version 16.11) in VS Code editor.

Step 1: Clone the Repository

```
git clone https://github.com/emmanuelnyandukagarabi/dpc_fam_and_struct_webapp
cd dpc_fam_and_struct_webapp
```

Step 2: Virtual Environment

```
python3 -m venv .area_internship_venv
source .area_internship_venv/bin/activate
```

Step 3: Install Project Dependencies

```
pip install -r requirements.txt
```

Step 4: Set Up PostgreSQL Database

```
# Installation
sudo apt update
```

```

sudo apt install postgresql postgresql-contrib

# Start
sudo service postgresql start

# Connect as postgres user:
sudo -u postgres psql

# Create a user
CREATE USER enyanduk WITH PASSWORD 'EmmaPSQL2026';

# Create the database
CREATE DATABASE dpcfam_mcs_db OWNER enyanduk;

# Grant all privileges
GRANT ALL PRIVILEGES ON DATABASE dpcfam_mcs_db TO enyanduk;

# Connect to the database and grant schema privileges
\c dpcfam_mcs_db

# Grant schema permissions
GRANT ALL ON SCHEMA public TO enyanduk;

# Exit
\q

# Verificatio (back in the terminal, not in psql: password = EmmaPSQL2026)
psql -U enyanduk -h localhost -d dpcfam_mcs_db -c "SELECT 1;"

```

2.2.1 Step 5: Configure Django Settings

```

# Database configuration : dpc_fam_and_struct_webapp/settings.py
DATABASES = {
    "default": {
        "ENGINE": "django.db.backends.postgresql",
        "NAME": "dpcfam_mcs_db",
        "USER": "enyanduk",
        "PASSWORD": "EmmaPSQL2026",
        "HOST": "localhost",
        "PORT": "5432",
    }
}

```

Step 6: Create Database Tables

```
# Navigate to the project and activate the virtual env and run a sql script
to create tables :
PGPASSWORD="EmmaPSQL2026" psql -U enyanduk -h localhost -d dpcfam_mcs_db
-f create_dpcfam_tables.sql

# Verify tables were created
PGPASSWORD="EmmaPSQL2026" psql -U enyanduk -h localhost -d dpcfam_mcs_db -c "\dt"

# We expect to see :
mcs_properties
mcs_sequences
alphafold_reps
```

Step 7: Import Data from CSV Files

```
# Download the data from One Drive :
# cd dpc_fam_and_struct_webapp/static/production_files/dpcfam/Tables/
wget https://areasciencepark-my.sharepoint.com/:f:/r/personal/emmanuel_nyandu_
kagarabi_areasciencepark_it/Documents/Attachments?csf=1&web=1&e=ecBNhd

# Unzip the folder ~3.5 GBs :
tar -xzf dpcfam_dataframes_report4.tar.gz

# Files should be:
dpcfam_mcs_props.csv
dpcfam_sequences.csv
alphafold_dpcfam_reps.csv

# venv must be activated to import data:

# Import MCS Properties (small file, fast)
python3 manage.py import_mcs_csv static/production_files/dpcfam/Tables/
dpcfam_mcs_props.csv

# Import Sequences (large file, takes time)
python3 manage.py import_mcs_sequences static/production_files/dpcfam/Tables/
dpcfam_sequences.csv

# Import AlphaFold Data (small file, fast)
python3 manage.py import_alphafold static/production_files/dpcfam/Tables/
alphafold_dpcfam_reps.csv
```

Step 8: Create Admin User

```
# Create a staff member(if needed :  
python3 manage.py createsuperuser  
# Follow the prompts and enter any name wanted, Mail and Password.
```

Step 9: Run the Development Server

```
python3 manage.py runserver
```

Step 10: Visit the Website

```
# Open a browser and land in the home page at  
http://localhost:8000
```

2.3 Current State

If the application of all previous steps was successful, the user may land on the following home page:

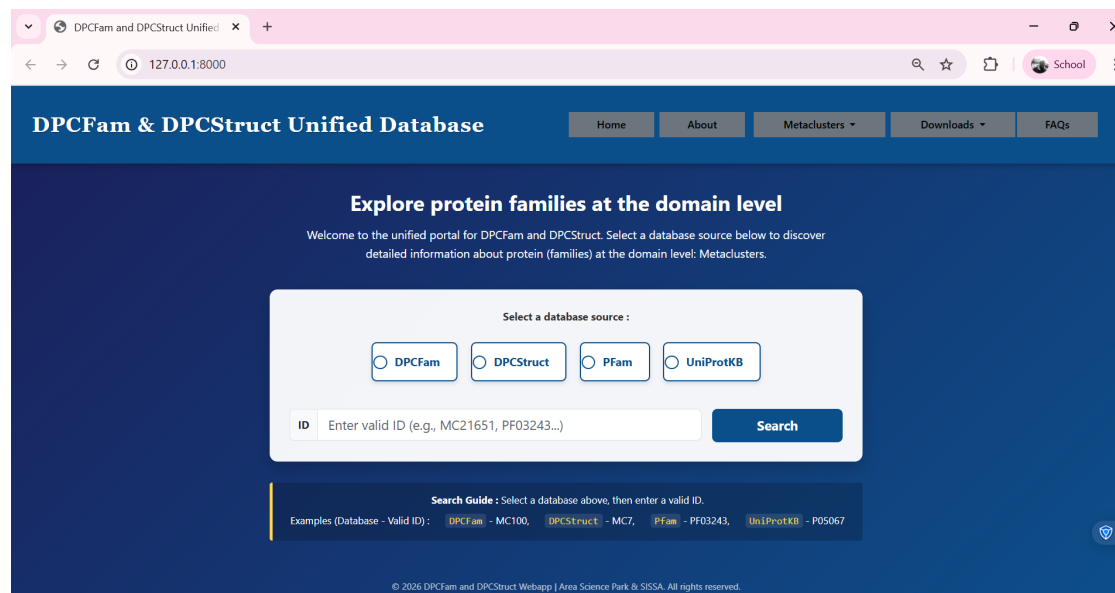


Figure 4: Landing Page.

The three created tables take the user respectively to :

127.0.0.1:8000/dpcfam/mcs/?page=1&mcid=MC1&sort=mcid

DPCFam & DPCStruct Unified Database

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Metaclusters

Enter MCID

McId	Size uniref50	Avg len	Std avg len	Lc percent	Cc percent	Dis percent	Tm	Pfam da	Da percent	Size pfam	Avg ov percent	Overlap label
MC1	17931	185.68	28.77	4.72	0.0	18.44	0.01	PF13614	44.23	6332	80.82	equivalent
MC100	81	107.57	13.13	0.62	0.0	17.4	0.0	UNKNOWN	0.0	61	0.0	NONE
MC100000	67	165.13	20.76	2.66	0.0	17.25	0.02	UNKNOWN	0.0	51	0.0	NONE
MC100002	66	82.06	6.13	0.91	0.39	11.42	0.0	UNKNOWN	0.0	55	0.0	NONE
MC100016	102	155.54	18.34	1.0	0.11	27.31	0.02	UNKNOWN	0.0	83	0.0	NONE
MC10002	122	155.74	20.3	1.25	0.0	14.71	0.0	PF08901	96.36	106	84.69	equivalent
MC10003	128	138.06	9.57	7.84	0.0	1.09	2.36	PF14296	1.12	1	40.6	shifted
MC100030	194	503.08	105.06	9.59	14.02	38.57	0.78	PF07738	99.45	181	25.27	extended
MC100031	106	108.42	15.45	1.09	0.64	17.43	0.0	PF05679	96.25	77	29.92	reduced

Figure 5: MCs properties paginated table.

127.0.0.1:8000/dpcfam/mcs/MC1/#sequences

DPCFam | MC1

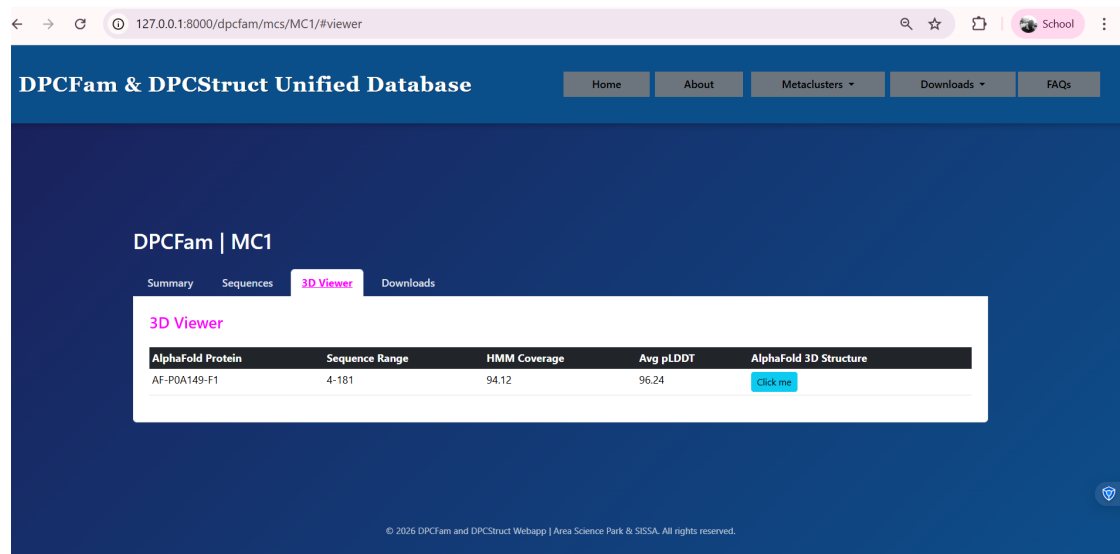
Summary **Sequences** 3D Viewer Downloads

Sequences

Protein ID	Range	Length	Amino Acids
A0A0M0COW3	3-192	190	SIIAVHSYKGGTGKTSLSVNLAA5LAKRGKVAI.LDLDFRAPSLFSLKPMEDSNLWINDYNGASTINKILQETNGKGLNKGKLF...
A0A0M0COW3	3-192	190	SIIAVHSYKGGTGKTSLSVNLAA5LAKRGKVAI.LDLDFRAPSLFSLKPMEDSNLWINDYNGASTINKILQETNGKGLNKGKLF...
A0A1P8WP13	533-701	169	VRTAVCF5ALGNAHRVIVQTS5PAAGDGKSTLAMNFAVSLAQSGKCTLIMESDFRRPKVHKLTGVDNKKGGVDVLRGKAEISDAIK...
A0A1P8WP13	533-701	169	VRTAVCF5ALGNAHRVIVQTS5PAAGDGKSTLAMNFAVSLAQSGKCTLIMESDFRRPKVHKLTGVDNKKGGVDVLRGKAEISDAIK...
A0A0P7YPQ9	114-308	195	PGVRTTLAVSSGGGVGKSTTAVNLAAL.TQTGAQVGLDADIVGPHVPQMLGLGHADVEIVKTAQDGRFKPLEAHGIQVMSVGL...
A0A1S9UN50	24-209	186	RSKISEQYRSIRTNLQLSASIHKNRTIVITSPRYGEGKSTITVNLAA5MAQKGEKLVVDANLRPTIIEHMFQVENTIGLTDILN...
A0A0P7YPQ9	114-308	195	PGVRTTLAVSSGGGVGKSTTAVNLAAL.TQTGAQVGLDADIVGPHVPQMLGLGHADVEIVKTAQDGRFKPLEAHGIQVMSVGL...
A0A094QKT1	2-116	115	KTITATASPAGGVGKTTLAHAISVAAAEFGKKTLLIDLPAGSLTFRLGFENPRLTITDFLTGTSLGDENLNLTAERFAFIPADSR...
A0A1S9UN50	24-209	186	RSKISEQYRSIRTNLQLSASIHKNRTIVITSPRYGEGKSTITVNLAA5MAQKGEKLVVDANLRPTIIEHMFQVENTIGLTDILN...
C1F829	251-474	224	VPVLAIFSVAGGVGKTSLTANLGRALSSYGECPLLAETSIHGVLPFYFGARTPRPGVLRFTFASHSAQEPVQMLSLDGRHIAEG...

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Figure 6: MCs Sequences paginated table.



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DPCFam | MC1

Summary Sequences **3D Viewer** Downloads

3D Viewer

AlphaFold Protein	Sequence Range	HMM Coverage	Avg pLDDT	AlphaFold 3D Structure
AF-P0A149-F1	4-181	94.12	96.24	Click me

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Figure 7: AlphaFold Representatives table.

3 Conclusion

To display this draft version, a local PostgreSQL database configured as described above is required. A bit more time is still needed to complete the DPCFam app properly before starting with DPCStruct.

References

- Russo, E. T., Barone, F., Bateman, A., Cozzini, S., Punta, M., and Laio, A. (2022). Dpcfam: unsupervised protein family classification by density peak clustering of large sequence datasets. *PLOS Computational Biology*, 18(10):e1010610.